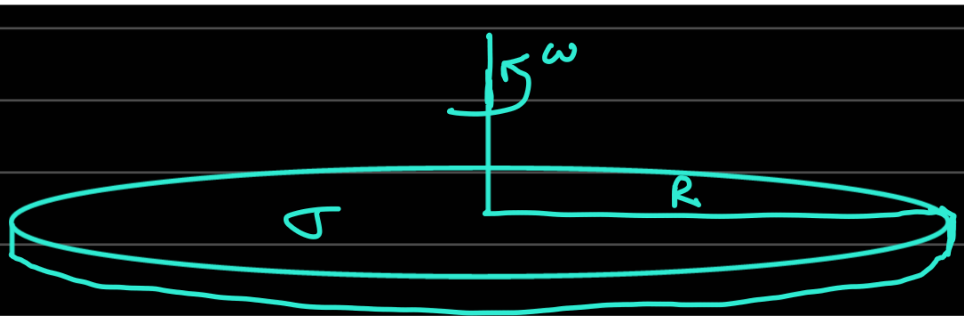
10. *Galaxy rotation curves and dark matter:* Many spiral galaxies are thin disks (with possibly a bulge and/or a bar in the middle), which rotate about an axis through their centre. Let the visible matter in the galaxy extend until radius R.

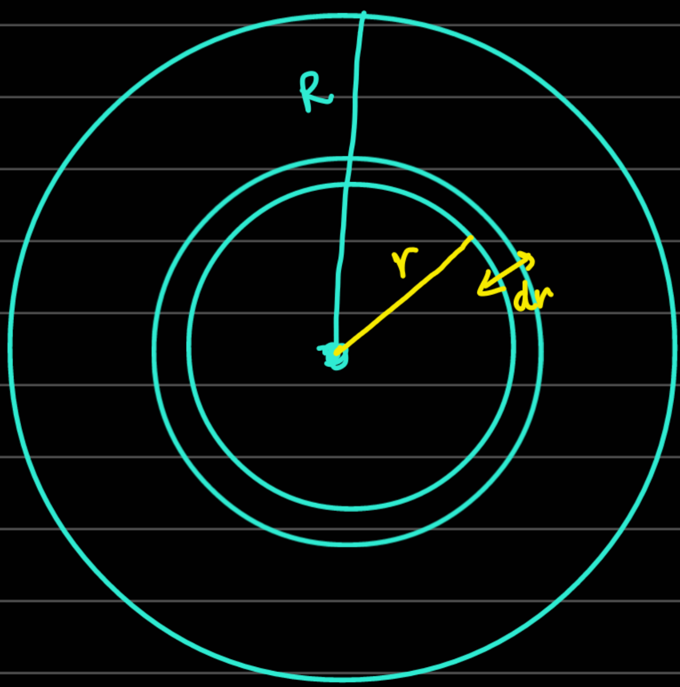
(a) Assuming a constant mass density until a radius R, determine the rotational velocity of the galaxy as a function of the radius r for $r < R$ and $r > R$.

(b) The so-called galaxy rotation curves of spiral galaxies indicate that the velocity is a constant for $r > R$. If so, assuming a simple power law for the density as a function of the radius, determine the radial dependence of the density of mass in the galaxy. What does the result imply?

Note: This suggests the existence of dark matter, i.e. matter that does not interact with light.

(c) If the rotation curves remain a constant until the radius of, say, $r = 8\,R$, estimate the fraction of mass of the galaxy that is in the form of dark matter?





for $r<R$,

$$\int dU = -\int \frac{G(\sigma\pi r^2)(\sigma 2\pi r)\,dr}{r}$$

$$U = -2G\sigma^2\pi^2 \int r^2\,dr$$

$$U = -\frac{2}{3}G\sigma^2\pi^2 r^3 = \langle U \rangle$$

$$dT = \frac{1}{2}\sigma 2\pi r\, dr\, \omega^2 r^2 = \sigma\pi\, \omega^2 r^3\, dr$$

$$\Rightarrow \quad T = \sigma\pi \int \omega^2 r^3\, dr = \langle T \rangle$$

From Virial theorem, $2\langle T\rangle = -\langle U\rangle$

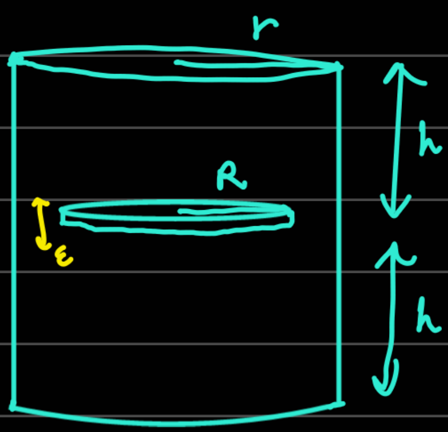
$$\Rightarrow 2\sigma\pi \int \omega^2 r^3\, dr = \frac{2}{3} G\sigma^2\pi^2 r^3$$

$$\int \omega^2 r^3\, dr = \frac{1}{3} G\sigma\pi r^3$$

$$\Rightarrow \quad \omega^2 r^3 = G\sigma\pi r^2$$

$$\boxed{\omega = \sqrt{\frac{G\sigma\pi}{r}}}$$

For $r > R$, use Gauss' Law.
Flux cancels everywhere except in plane of disc (radially outward field)



WTF

1. *Motion in one dimension:* Consider a particle that is in motion in the following one-dimensional potential:

$$U(x) = \alpha\left(e^{-2\beta x} - 2e^{-\beta x}\right), \text{ where } (\alpha, \beta) > 0.$$

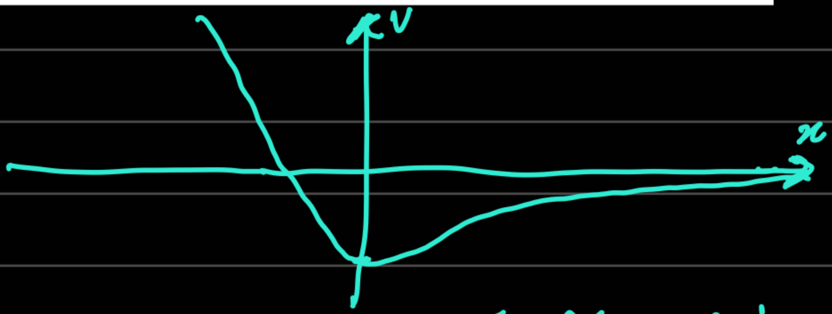
Let E be the energy of the particle.

(a) Obtain the solutions describing the time evolution of the particle for the cases $E < 0$, $E = 0$ and $E > 0$.

(b) Evaluate the period of oscillation of the particle when $E < 0$.

1

a) $E = \frac{1}{2}m\dot{x}^2 + U$



$U(x) = \alpha\left(e^{-2\beta x} - 2e^{-\beta x}\right)$

$\Rightarrow \sqrt{\frac{2}{m}(E-U)} = \frac{dx}{dt}$

$\Rightarrow \quad dt = \sqrt{\frac{m}{2}}\frac{dx}{\sqrt{E-U}}$

For $E<0$, let $E = -\varepsilon$, $\varepsilon > 0$.

$$dt = \sqrt{\frac{m}{2}}\frac{dx}{\sqrt{-\varepsilon + 2\alpha e^{-\beta x} - \alpha e^{-2\beta x}}}$$

$$dt = \sqrt{\frac{m}{2}}\frac{e^{\beta x}\,dx}{\sqrt{-\varepsilon e^{2\beta x} + 2\alpha e^{\beta x} - \alpha}}$$

let $u = e^{\beta x} \Rightarrow du = \beta e^{\beta x}\,dx$

$$dt = \frac{1}{\beta}\sqrt{\frac{m}{2}}\frac{du}{\sqrt{-\varepsilon u^2 + 2\alpha u - \alpha}}$$

$$-\varepsilon u^2 + 2\alpha u - \alpha = -\varepsilon\left(u^2 - \frac{2\alpha}{\varepsilon}u + \frac{\alpha}{\varepsilon}\right)$$

$$= -\varepsilon\left[\left(u - \frac{\alpha}{\varepsilon}\right)^2 + \frac{\alpha}{\varepsilon} - \frac{\alpha^2}{\varepsilon^2}\right]$$

$$= \varepsilon\left(\frac{\alpha^2}{\varepsilon^2} - \frac{\alpha}{\varepsilon} - \left(u - \frac{\alpha}{\varepsilon}\right)^2\right)$$

$$\int dt = \frac{1}{\beta}\sqrt{\frac{m}{2\varepsilon}} \int \frac{du}{\sqrt{A^2 - \left(u - \frac{\alpha}{\varepsilon}\right)^2}} \qquad A^2 = \frac{\alpha^2}{\varepsilon^2} - \frac{\alpha}{\varepsilon}$$

$$u - \frac{\alpha}{\varepsilon} = A\cos\theta \quad \Rightarrow \quad du = -A\sin\theta \, d\theta$$

$$t - t_0 = \frac{-1}{\beta}\sqrt{\frac{m}{2\varepsilon}} \cos^{-1}\left(\frac{u}{A} - \frac{\alpha}{A\varepsilon}\right)$$

$$\frac{\alpha}{\varepsilon} + A\cos\left(\beta\sqrt{\frac{2\varepsilon}{m}}(t-t_0)\right) = e^{\beta x}$$

$$\Rightarrow \quad \boxed{x = \frac{1}{\beta}\ln\left(\frac{\alpha}{\varepsilon} + A\cos\left(\beta\sqrt{\frac{2\varepsilon}{m}}(t-t_0)\right)\right)}$$

where $A^2 = \frac{\alpha^2}{\varepsilon^2} - \frac{\alpha}{\varepsilon}$

$\varepsilon = |E|$

For $E = 0$,

$$dt = \sqrt{\frac{m}{2}} \frac{dx}{\sqrt{-U}}$$

$$dt = \sqrt{\frac{m}{2\alpha}} \frac{dx}{\sqrt{\left(2e^{-\beta x} - e^{-2\beta x}\right)}}$$

$$dt = \sqrt{\frac{m}{2\alpha}} \frac{e^{\beta x}\, dx}{\sqrt{2e^{\beta x} - 1}}$$

$$u^2 = 2e^{\beta x} - 1 \quad \Rightarrow \quad u\,du = \beta e^{\beta x} dx$$

$$dt = \frac{1}{\beta}\sqrt{\frac{m}{2\alpha}}\, du$$

$$t - t_0 = \frac{1}{\beta}\sqrt{\frac{m}{2\alpha}} \sqrt{2e^{\beta x} - 1}$$

$$\frac{\alpha\beta^2}{m}(t-t_0)^2 + \frac{1}{2} = e^{\beta x}$$

$$\Rightarrow \boxed{x = \frac{1}{\beta} \ln\left(\frac{1}{2} + \frac{\alpha\beta^2}{m}(t-t_0)^2\right)}$$

For $E > 0$,

$$dt = \sqrt{\frac{m}{2}} \frac{dx}{\sqrt{E-U}}$$

$$dt = \sqrt{\frac{m}{2}} \frac{dx}{\sqrt{E - \alpha e^{-2\beta x} + 2\alpha e^{-\beta x}}}$$

$$dt = \sqrt{\frac{m}{2}} \frac{e^{\beta x}\, dx}{\sqrt{E e^{2\beta x} + 2\alpha e^{\beta x} - \alpha}}$$

$$dt = \sqrt{\frac{m}{2E}} \frac{e^{\beta x}\, dx}{\sqrt{e^{2\beta x} + \frac{2\alpha}{E} e^{\beta x} - \frac{\alpha}{E}}}$$

$$u = e^{\beta x} \Rightarrow du = \beta e^{\beta x} dx$$

$$dt = \frac{1}{\beta}\sqrt{\frac{m}{2E}} \frac{du}{\sqrt{u^2 + \frac{2\alpha}{E} u - \frac{\alpha}{E}}}$$

$$dt = \frac{1}{\beta}\sqrt{\frac{m}{2E}} \frac{du}{\sqrt{\left(u + \frac{\alpha}{E}\right)^2 - \left(\frac{\alpha^2}{E^2} + \frac{\alpha}{E}\right)}}$$

Let $A^2 = \frac{\alpha^2}{E^2} + \frac{\alpha}{E}$ and $u + \frac{\alpha}{E} = A \cosh \xi$

$$\int dt = \frac{1}{\beta}\sqrt{\frac{m}{2E}} \int d\xi$$

$$t - t_0 = \frac{1}{\beta}\sqrt{\frac{m}{2E}} \cosh^{-1}\left(\frac{u}{A} + \frac{\alpha}{AE}\right)$$

$$A\cosh\left(\beta\sqrt{\frac{2E}{m}}(t-t_0)\right) - \frac{\alpha}{E} = e^{\beta x}$$

$$\Rightarrow \boxed{x = \frac{1}{\beta}\ln\left(A\cosh\left(\beta\sqrt{\frac{2E}{m}}(t-t_0)\right) - \frac{\alpha}{E}\right)}$$

where $A^2 = \frac{\alpha^2}{E^2} + \frac{\alpha}{E}$

b)

$$\boxed{x = \frac{1}{\beta}\ln\left(\frac{\alpha}{\varepsilon} + A\cos\left(\beta\sqrt{\frac{2\varepsilon}{m}}(t-t_0)\right)\right)}$$

$$E = \alpha\left(e^{-2\beta x} - 2e^{-\beta x}\right) \qquad A = \sqrt{\frac{\alpha^2}{\varepsilon^2} - \frac{\alpha}{\varepsilon}}$$

$$\frac{E}{\alpha}e^{2\beta x} = 1 - 2e^{\beta x} \qquad \frac{\alpha}{\varepsilon}\left(1-\frac{\alpha}{\varepsilon}\right)$$

$$\frac{E}{\alpha}e^{2\beta x} + 2e^{\beta x} - 1 = 0$$

$$e^{\beta x} = \frac{-2 \pm \sqrt{4 + \frac{4E}{\alpha}}}{\frac{2E}{\alpha}}$$

$$e^{\beta x} = \frac{-\alpha \pm \sqrt{\alpha^2 + \alpha E}}{E}$$

But $E = -\varepsilon$

$$e^{\beta x} = \frac{\alpha \pm \sqrt{\alpha^2 - \alpha\varepsilon}}{\varepsilon}$$

$$x_1 = \frac{1}{\beta}\ln\left(\frac{\alpha + \sqrt{\alpha^2 - \alpha\varepsilon}}{\varepsilon}\right), \quad x_2 = \frac{1}{\beta}\ln\left(\frac{\alpha - \sqrt{\alpha^2 - \alpha\varepsilon}}{\varepsilon}\right)$$

So $A\cos\left(\beta\sqrt{\frac{2\varepsilon}{m}}(t-t_0)\right) = \frac{\sqrt{\alpha^2 - \alpha\varepsilon}}{\varepsilon}$ and $-\frac{\sqrt{\alpha^2 - \alpha\varepsilon}}{\varepsilon}$

$$= \sqrt{\frac{\alpha^2}{\varepsilon^2} - \frac{\alpha}{\varepsilon}} \quad \text{and} \quad -\sqrt{\frac{\alpha^2}{\varepsilon^2} - \frac{\alpha}{\varepsilon}}$$

$$= A \quad \text{and} \quad -A$$

$$\Rightarrow \cos\left(\beta\sqrt{\frac{2\mathcal{E}}{m}}\,(t-t_0)\right) = 1 \quad \text{and} \quad -1$$

$$\Rightarrow \beta\sqrt{\frac{2\mathcal{E}}{m}}\,(t-t_0) = 0 \quad \text{and} \quad \beta\sqrt{\frac{2\mathcal{E}}{m}}\,(t-t_0) = \pi$$

$$t = t_0 \qquad\qquad t = t_0 + \frac{\pi}{\beta}\sqrt{\frac{m}{2\mathcal{E}}}$$

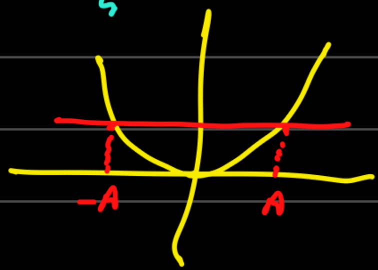
$$\therefore \frac{T}{2} = \frac{\pi}{\beta}\sqrt{\frac{m}{2\mathcal{E}}} \Rightarrow \boxed{T = \frac{2\pi}{\beta}\sqrt{\frac{-m}{2E}}}$$

2. <u>*Time period of the Duffing's oscillator:*</u> A Duffing oscillator is a non-linear oscillator that is governed by the potential

$$U(x) = \frac{1}{2}\,m\,\omega^2\,x^2 + \frac{1}{4}\,\epsilon\,\alpha\,x^4,$$

where ϵ is a dimensionless, infinitesimal and positive quantity, while $\alpha > 0$ has suitable dimensions. Express the time period of the oscillator in terms of its amplitude, say, A, at the order ϵ.

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\omega^2x^2 + \frac{1}{4}\epsilon\alpha x^4 = \frac{1}{2}m\omega^2A^2 + \frac{1}{4}\epsilon\alpha A^4$$



$$E - \frac{1}{2}m\omega^2x^2 - \frac{1}{4}\epsilon\alpha x^4 = \sqrt{\frac{m}{2}}\,\frac{dx}{dt}$$

$$\Rightarrow dt = \sqrt{\frac{m}{2}}\,\frac{dx}{\sqrt{E - \frac{1}{2}m\omega^2x^2 - \frac{1}{4}\epsilon\alpha x^4}}$$

$$dt = \sqrt{\frac{m}{2}}\,\frac{dx}{\sqrt{\frac{1}{4}\epsilon\alpha(A^4 - x^4) + \frac{1}{2}m\omega^2(A^2 - x^2)}}$$

$$dt = \sqrt{\frac{m}{2}}\,\frac{dx}{\sqrt{\frac{1}{2}m\omega^2(A^2-x^2)}\sqrt{\frac{\epsilon\alpha}{m\omega^2}(A^2+x^2) + 1}}$$

$$dt = \sqrt{\frac{m}{2}}\,\frac{\left[1 - \frac{\epsilon\alpha}{2m\omega^2}(A^2+x^2)\right]}{\sqrt{\frac{1}{2}m\omega^2(A^2-x^2)}}\,dx$$

$$\int_0^{T/2} dt = \sqrt{\frac{m}{2}}\int \frac{dx}{\sqrt{\frac{1}{2}m\omega^2(A^2-x^2)}} - \frac{\varepsilon\alpha}{2m\omega^2}\sqrt{\frac{m}{2}}\int \frac{A^2+x^2}{\sqrt{\frac{1}{2}m\omega^2(A^2-x^2)}}\,dx$$

$$\frac{T}{2} = \frac{T_0}{2} - \frac{\varepsilon\alpha}{2m\omega^3}\int_{-A}^{A} \frac{A^2+x^2}{\sqrt{A^2-x^2}}\,dx$$

$$\frac{T}{2} = \frac{T_0}{2} - \frac{\varepsilon\alpha A}{2m\omega^3}\int_{-A}^{A} \frac{1+x^2/A^2}{\sqrt{1-x^2/A^2}}\,dx$$

$$x = A\sin\theta \qquad dx = A\cos\theta\, d\theta$$

$$\frac{T}{2} = \frac{T_0}{2} - \frac{\varepsilon\alpha A^2}{2m\omega^3}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1+\sin^2\theta)\,d\theta$$

$$\frac{T}{2} = \frac{T_0}{2} - \frac{3\varepsilon\alpha A^2\pi}{4m\omega^3} \Rightarrow T = T_0 - \frac{3\varepsilon\alpha A^2}{4m\omega^2}\,\frac{2\pi}{\omega}$$

$$\boxed{T = T_0\left(1 - \frac{3\varepsilon\alpha A^2}{4m\omega^2}\right)}$$

3. *Zero potential:* Solve the orbital equation when the potential is zero. What is the shape of the trajectory?

$$E = \frac{1}{2}m\dot{r}^2 + \frac{L^2}{2mr^2}$$

$$E = \left(\frac{1}{r^2}\frac{dr}{d\phi}\right)^2\frac{L^2}{2m} + \frac{L^2}{2mr^2}$$

$$u = 1/r \quad , \quad \frac{du}{d\phi} = -\frac{1}{r^2}\frac{dr}{d\phi}$$

$$E = \left(\frac{du}{d\phi}\right)^2\frac{L^2}{2m} + \frac{L^2u^2}{2m}$$

$$\sqrt{\frac{2mE}{L^2} - u^2} = \frac{du}{d\phi}$$

$$d\phi = \frac{du}{\sqrt{\alpha^2 - u^2}}$$

$$\alpha^2 = \frac{2mE}{L^2}$$

$$\alpha \sin(\phi - \phi_0) = \frac{1}{r} \Rightarrow \boxed{r = \frac{1}{\alpha \sin(\phi - \phi_0)}}$$

Straight lines

4. *Two massive bodies in a constant field:* Two bodies move in a constant external gravitational field. Show that their motion can be reduced to an equivalent one body problem, as in the case of the Kepler problem.

$\bar{r}_1 : m_1$
$\bar{r}_2 : m_2$

$V_1(r) = K_1 r_1$
$V_2(r) = K_2 r_2$

$$\mathcal{L} = \frac{1}{2} m_1 \dot{\bar{r}}_1^2 + \frac{1}{2} m_2 \dot{\bar{r}}_2^2 - K_1 r_1 - K_2 r_2 - U(|\bar{r}_1 - \bar{r}_2|)$$

$$m_1 \bar{r}_1 + m_2 \bar{r}_2 = 0$$
$$\bar{r} = \bar{r}_1 - \bar{r}_2 \Rightarrow \bar{r}_1 = \bar{r} - \frac{m_1}{m_2}\bar{r}_1 \Rightarrow \bar{r}_1 = \frac{m_2}{m_1+m_2}\bar{r}$$

$$\bar{r}_2 = \frac{-m_1 \bar{r}}{m_1+m_2}$$

$$\mathcal{L} = \frac{1}{2} m_1 \frac{m_2^2 \dot{r}^2}{(m_1+m_2)^2} + \frac{1}{2} m_2 \frac{m_1^2 \dot{r}^2}{(m_1+m_2)^2} - \frac{K_1 m_2}{m_1+m_2} r + \frac{K_2 m_1}{m_1+m_2} r - U(r)$$

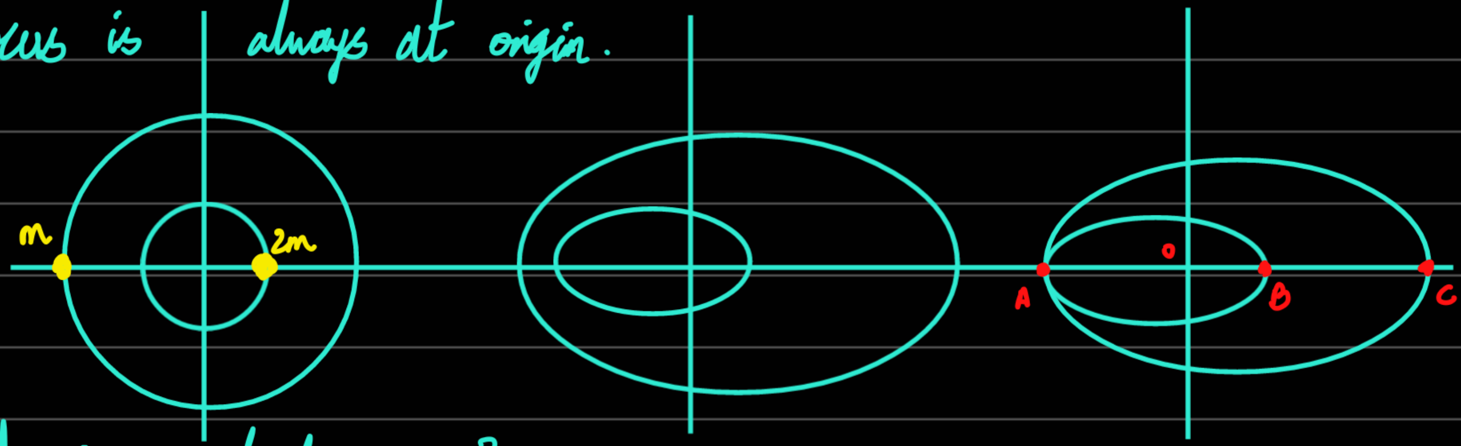
$$\boxed{\mathcal{L} = \frac{1}{2}\mu \dot{r}^2 + K' r - U(r)}$$

and if $K \propto m$ (as it is a gravitational field, then $K' = 0$)

where $K' = \frac{K_2 m_1 - K_1 m_2}{m_1+m_2}$, $\mu = \frac{m_1 m_2}{m_1+m_2}$

5. *Intersecting orbits:* Two masses, M and $2\,M$, orbit around their centre of mass. If the orbits are circular, they do not intersect. But if they are very elliptical, they do. What is the smallest value of the eccentricity for which they intersect?

Focus is always at origin.



Why is eccentricity same?

① $- \quad 2m\, a_1(1-e_1) = m\, a_2(1-e_2)$ for CoM to be at O when 2m at B and m at A

② $- \quad 2m\, a_1(1+e_1) = m\, a_2(1+e_2)$ for CoM to be at O when 2m at A and m at C.

$$\Rightarrow \quad \frac{1+e_1}{1-e_1} = \frac{1+e_2}{1-e_2} \Rightarrow e_1 = e_2 = e \quad (\text{let})$$

So from 1, $\quad \dfrac{a_1}{a_2} = \dfrac{1}{2}$

Also $OA = a_1(1+e) = a_2(1-e) \Rightarrow 1+e = 2-2e$

$$\Rightarrow \boxed{e = \frac{1}{3}}$$

6. *Potentials permitting closed orbits:* Obtain the condition on attractive power law central potentials if the orbits in the potentials are to remain closed.

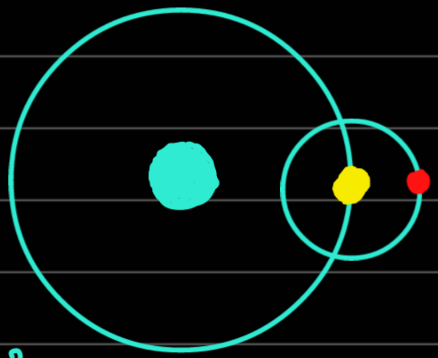
Note: The result is known as Bertrand's theorem.

Too irrelevant.

8. *Ratio of the mass of the Sun to the Earth:* Evaluate, approximately, the ratio of the mass of the Sun to that of the Earth, using only the lengths of the year and of the lunar month (which is 27.3 days) and the mean radii of the orbits of the Earth and the Moon, which are 1.49×10^8 km and 3.8×10^5 km, respectively.

$$T_1^2 = \frac{4\pi^2 R_1^3}{GM}$$

$$T_2^2 = \frac{4\pi^2}{GM_e} R_2^3$$



$$\Rightarrow \frac{T_1^2}{T_2^2} = \frac{M_e}{M} \frac{R_1^3}{R_2^3} \Rightarrow \frac{M}{M_e} = \frac{T_2^2}{T_1^2} \frac{R_1^3}{R_2^3}$$

$$\approx \boxed{3.372 \times 10^5}$$

9. *Return of comets Hyakutake and Hale-Bopp:* The comets Hyakutake and Hale-Bopp appeared in March–May 1996 and March–May 1997. If their eccentricities and perihelion are $e = 0.999846$ and 0.995075 and 0.230123 AU and 0.913959 AU, respectively, determine when the comets will be seen again.

$$a_1 = \frac{r_1}{1-e_1} \qquad a_2 = \frac{r_2}{1-e_2}$$

$M \to$ sun

$$T_1^2 = \frac{4\pi^2}{GM}\left(\frac{r_1}{1-e_1}\right)^3 \qquad T_2^3 = \frac{4\pi^2}{GM}\left(\frac{r_2}{1-e_2}\right)^3$$

Now say Earth makes n revolutions and comet makes m revolutions before they meet again. Then

$$n T_{Earth} = m T_i$$

$$\Rightarrow \frac{n}{m} = \frac{2\pi a_{comet}^{3/2}}{\sqrt{GM}} \cdot \frac{\sqrt{GM}}{2\pi a_{earth}^{3/2}}$$

$$\gamma = \frac{n}{m} = \left(\frac{a_{comet}}{a_{earth}}\right)^{3/2} \qquad (a_{earth} = 1 \text{ AU})$$

$$\Rightarrow \gamma_1 = 57764 \text{ and } \gamma_2 = 2528$$

So Hyakutake returns after 57,764 years and Hale-Bopp after 2528 years.

7. *The isotropic oscillator:* Consider the isotropic oscillator described by the potential

$$U(r) = \frac{1}{2} m\omega^2 r^2.$$

(a) Solve the corresponding orbital equation.

(b) What is the shape of the general trajectory?

(c) Also, solve for the time dependence of the radial and the angular coordinates.

$$E = \frac{l^2}{2m}\left(\frac{1}{r^2}\frac{dr}{d\phi}\right)^2 + \frac{l^2}{2mr^2} + \frac{1}{2}m\omega^2 r^2$$

$$E - \frac{l^2 u^2}{2m} - \frac{m\omega^2}{2u^2} = \frac{l^2}{2m}\left(\frac{du}{d\phi}\right)^2$$

$$\sqrt{\frac{2mE}{l^2} - u^2 - \frac{m^2\omega^2}{l^2 u^2}} = \frac{du}{d\phi}$$

$$\Rightarrow \quad d\phi = \frac{du}{\sqrt{\frac{2mE}{l^2} - \left(\frac{m^2\omega^2}{l^2u^2} + u^2\right)}}$$

$$d\phi = \frac{u\,du}{\sqrt{\frac{2mE}{l^2}u^2 - \frac{m^2\omega^2}{l^2} - u^4}}$$

$$s = u^2$$

$$\frac{ds}{2} = u\,du$$

$$d\phi = \frac{ds/2}{\sqrt{-s^2 + \frac{2mE}{l^2}s - \frac{m^2\omega^2}{l^2}}}$$

$$-\left(s^2 - \frac{2mE}{l^2}s + \frac{m^2\omega^2}{l^2}\right) = -\left(s - \frac{mE}{l^2}\right)^2 - \frac{m^2\omega^2}{l^2} + \frac{m^2E^2}{l^4}$$

$$d\phi = \frac{ds/2}{\sqrt{\underbrace{\left(\frac{m^2E^2}{l^4} - \frac{m^2\omega^2}{l^2}\right)}_{A^2} - \left(s - \frac{mE}{l^2}\right)^2}}$$

$$s - \frac{mE}{l^2} = A \sin p$$

$$ds = A \cos p \; dp$$

$$d\phi = \frac{1}{2}\, dp$$

$$\sin\left(2(\phi - \phi_0)\right) = \frac{s}{A} - \frac{mE}{A l^2}$$

$$\frac{1}{r^2} = \frac{mE}{l^2} + A \sin\left(2(\phi - \phi_0)\right)$$

$$\frac{1}{r^2} = \frac{mE}{l^2} + \sqrt{\frac{m^2E^2}{l^4} - \frac{m^2\omega^2}{l^2}}\; \sin\left(2(\phi - \phi_0)\right)$$

$$\frac{1}{r^2} = \frac{mE}{l^2}\left(1 + \sqrt{1 - \frac{\omega^2 l^2}{E^2}}\; \sin 2\phi\right) \qquad (\phi_0 = 0)$$

$$\Rightarrow \quad \boxed{r^2 = \frac{q}{1 + f \sin 2\phi}} \qquad q = l^2/mE, \quad f = \sqrt{1 - \frac{\omega^2 l^2}{E^2}}$$

These are ellipses.

Now, $$dt = \sqrt{\frac{m}{2}}\; \frac{dr}{\sqrt{E - \frac{1}{2}m\omega^2 r^2 - \frac{L^2}{2mr^2}}}$$

$$dt = \sqrt{\frac{m}{2}} \frac{r\,dr}{\sqrt{-\frac{1}{2}m\omega^2 r^4 + Er^2 - \frac{L^2}{2m}}}$$

$$r^2 = u$$

$$\Rightarrow r\,dr = du/2$$

$$dt = \sqrt{\frac{m}{2}} \frac{du/2}{\sqrt{-\frac{1}{2}m\omega^2 u^2 + Eu - \frac{L^2}{2m}}}$$

$$-\left(\frac{1}{2}m\omega^2 u^2 - Eu + \frac{L^2}{2m}\right) = -\frac{1}{2}m\omega^2\left(u^2 - \frac{2E}{m\omega^2}u + \frac{L^2}{m^2\omega^2}\right)$$

$$= \frac{1}{2}m\omega^2\left(-\left(u - \frac{E}{m\omega^2}\right)^2 - \frac{L^2}{m^2\omega^2} + \frac{E^2}{m^2\omega^4}\right)$$

$$= \frac{E^2}{2m\omega^2} - \frac{L^2}{2m} - \frac{1}{2}m\omega^2\left(u - \frac{E}{m\omega^2}\right)^2$$

$$dt = \frac{1}{2\omega} \frac{du}{\sqrt{\frac{E^2}{m^2\omega^4} - \frac{L^2}{m^2\omega^2} - \left(u - \frac{E}{m\omega^2}\right)^2}}$$

But $\frac{E^2}{m^2\omega^4} - \frac{L^2}{m^2\omega^2}$

$$= \frac{E^2}{m^2\omega^4}\left(1 - \frac{\omega^2 L^2}{E^2}\right) = \frac{E^2 f^2}{m^2\omega^4} = B^2$$

$$dt = \frac{1}{2\omega B} \frac{du}{\sqrt{1 - \frac{\left(u - \frac{E}{m\omega^2}\right)^2}{B^2}}}$$

$$u - \frac{E}{m\omega^2} = B\sin s \quad \Rightarrow \quad du = B\cos s\, ds$$

$$dt = \frac{1}{2\omega}\, ds \Rightarrow \sin\left(2\omega(t-t_0)\right) = \frac{u}{B} - \frac{E}{mB\omega^2}$$

$$\Rightarrow \boxed{r^2 = \frac{E}{m\omega^2} + B\sin\left(2\omega(t-t_0)\right)}$$

where $B = \dfrac{Ef}{m\omega^2}$

$$f = \sqrt{1 - \frac{\omega^2 l^2}{E^2}}$$